Scientific African 22 (2023) e01913

Contents lists available at ScienceDirect

Scientific African

journal homepage: www.elsevier.com/locate/sciaf

Optimal design and sizing of a multi-microgrids system: Case study of Goma in The Democratic Republic of the Congo

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ARTICLE INFO

Editor: DR B Gyampoh

Keywords:
Microgrid
Multi-microgrids
Power quality
Phase-locked loop controller
Distributed renewable energy

ABSTRACT

Nowadays, renewable energy generation has gained a lot of attention in the global electrical energy production. This has led to an increased integration of distributed generators into the traditional electrical grid, standalone generations as well as the formation of microgrids. The continuous increase in microgrid penetration has resulted in the microgrids becoming neighbours to one another, thus, favouring the interconnection of these microgrids to form the multi-microgrids. The multi-microgrids infrastructure has the potential to improve energy availability, reliability and reduction in the net cost of energy through sharing of resources. The notion of unidirectional power flow that normally applies to the traditional electrical grid is no longer applicable in this context. This paper investigates the advantages of several microgrids' interconnection on the system reliability within the town of Goma in the Democratic Republic of the Congo (DRC) using the Homer Grid software for optimal sizing of components considering technical and economic aspects. From the potential assessment of this case study, it is realized that the methane gas present in Kivu Lake and solar PV are potential distributed sources that can be used. In addition, it was observed that when connected to the Virunga private grid, the system with PV can be sold to the grid with a payback period of about seven years. The system made of Generator, PV and battery was found to be the most reliable. This led to the conclusion that, the interconnection of several microgrids produces a more stable and reliable system, especially when different sources of energy are used.

Introduction

Electrical Energy is one of the most essential needs for the development of a community. The growing demand for energy and climate change concerns have led to increased interest in the use of renewable energy resources (RES) for technological and cooking applications [1,2]. Over 70% of global electricity production comes from non-renewable energy sources (such as fossil fuel: coal (38.5%), natural gas (23.0%), oil (3.3%) and, nuclear (10.3%)) [3] The use of these conventional energy sources is not environmentally friendly, resulting in the emission of greenhouse gases which contribute to global warming and climate change [4]. In an attempt to limit global warming and protect the planet, high integration of renewable energy sources (RES) in the electricity grid is neede [5] This has led to an increased integration of distributed generation (DG) units in the grid, and subsequently to the formation of microgrids constituted predominantly by renewable energy sources [6]. A major drawback to the integration of RES in microgrids is

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https://doi.org/10.1016/j.sciaf.2023.e01913
Received 19 May 2023; Received in revised form 12 July 2023; Accepted 20 September 2023
Available online 23 September 2023

the intermittent generation nature of most renewable energy sources (such as wind and solar), design challenges as well the high cost of energy from these sources. Different optimization techniques have been used to solve most the design problems using hybrid renewable energy systems resulting to improved reliability, yield and lowering the cost of energy [6–9].

Power quality issues in renewable energy sources used in microgrid systems

Unlike the traditional electrical grid, the generation of energy in a microgrid is based mainly on clean renewable energy sources such as wind and solar. However, due to their intermittent nature, there is imbalance between production and consumption. Besides this, the use of power electronics devices in DG introduces harmonics in the grid. Hence, the main challenges to deal with within a microgrid are:

Ø The need for huge energy storage system [10]
Ø Power quality issues [10–12].

More explicitly, common power quality and power sharing issues in microgrids introduced by distributed generation units are:

- Voltage regulation and voltage sag
- Voltage distortion
- Voltage imbalance
- Frequency variation
- Sustained interruption
- Bidirectional power flow.

These issues have as consequences the malfunction of protective relays, overheating of motors and transformers and failure of power factor correction capacitors.

Purpose

The widespread emergence of microgrids integrating RES has made microgrids neighbors to one another in recent years. These microgrids can share their power, and a multi-microgrid formed, which is a more stable and flexible mini grid. The above challenges found in a single microgrid become even more complex in the context of a multi-microgrid [13,14]. In trying to mitigate these challenges and to improve the returns on investment, this research work is focused on investigating the influence of multi-microgrids interconnection on reliability and techno-economic benefits via the optimal design and sizing of the multi-microgrids.

Methodology

Presentation of different procedures of the multi-microgrids design

As shown in on Fig. 1, the design of a multi-microgrids system was performed in three main parts:

1. Since the multi-microgrid contains several microgrids, we started by designing each microgrid present in the system,
2. The microdrids were then interconnected to form the multi-microgrids,
3. The multi-microgrids was then optimized to obtain the most viable configuration.

While designing a system using the Homer Grid software, the first step is to determine the load profile of the concerned microgrid in order to allow the software to optimize the size of all system's components based on technical and economic considerations.

The energy potential assessment of Goma

The case study of this research is the electrical network of the Goma town in the Democratic Republic of the Congo. Fig. 2 presents the aerial view of the chosen case study with its geographic coordinates (1^{0}39.5′S, 29^{0}13.2′E).

The Renewable energy potential of Goma in DRC is presented by Figs. 3 and 4. In these Figures we can see that the town does not

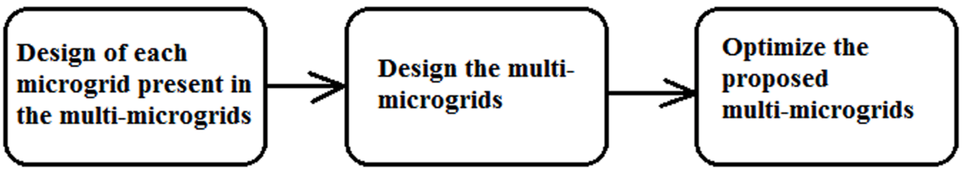

Fig. 1. Block diagram of the multi-microgrids design steps.

Fig. 2. Presentation of the aerial view of Goma town in the Democratic Republic of the Congo (DRC).

have enough wind potential to be used in electrical power production (comparing Fig. 4 and Fig. 5). On the other hand, the solar potential is relatively high.

Apart from the wind and solar energy resources discussed above, Goma is bordered on the south by the Kivu Lake that contains a high quantity of methane gas: 60 billion metre cube with the capacity to produce between 120 and 250 million cubic meters annually [17]. These alternative energy sources can be integrated into different microgrids configurations

Electrical demand of Goma

It is estimated that Goma requires 80 megawatts of electricity to accommodate the needs of residents and businesses. But to date, only 16.5 MW is available. The national grid (Société National d'Electricité) SNEL's capacity is 6.5 MW, while the private companies SOCODEE and Virunga, supply about 5 MW each [18].

System architecture

The multi-microgrids studied is composed of four microgrids interconnected at the medium voltage level (15 kV) through a transformer as shown in Fig. 6. The first microgrid has photovoltaic sources, batteries for energy storage and AC load. A converter is used for energy transfer between the DC and the AC bus. This microgrid operates at 400 V. The second microgrid has a methane gas generator and a 3 kV AC load. The third contains a 3 kV methane gas generator, PV sources, batteries, a bidirectional DC-AC converter and an AC load. And the last microgrid is based on PV and wind generation units working at the voltage of 400 V. The wind turbine is included here in order to validate, by Homer Grid optimization, the feasibility of wind energy in the town of Goma.

Results and discussion of the multi-microgrid design

Load profiles' presentation of the different microgrids configurations

Figs. 7–10 present the load profiles for microgrid 1, microgrid 2, microgrid 3 and microgrid 4 respectively.

It can be observed from the results shown on Figs. 7–10 that, the optimal wind turbine installed power is zero watt. This confirms the observation made earlier that the town of Goma does not have wind energy potentials. On the other hand, we can see that, it was

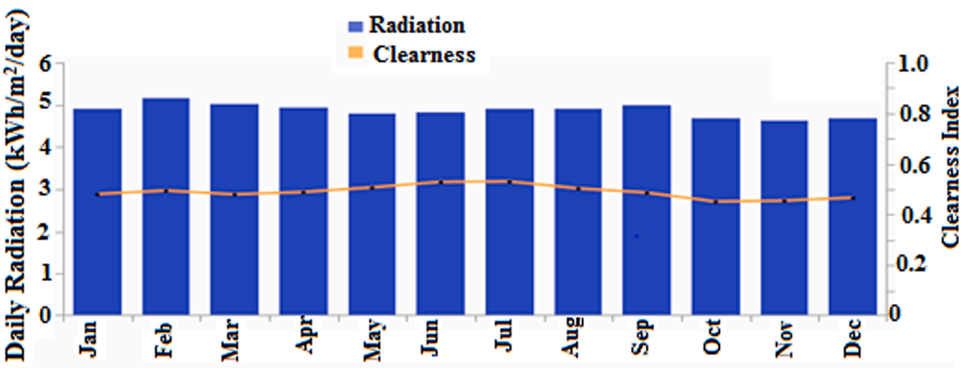

Fig. 3. Solar potential of Goma, obtained from SODA web site [15].

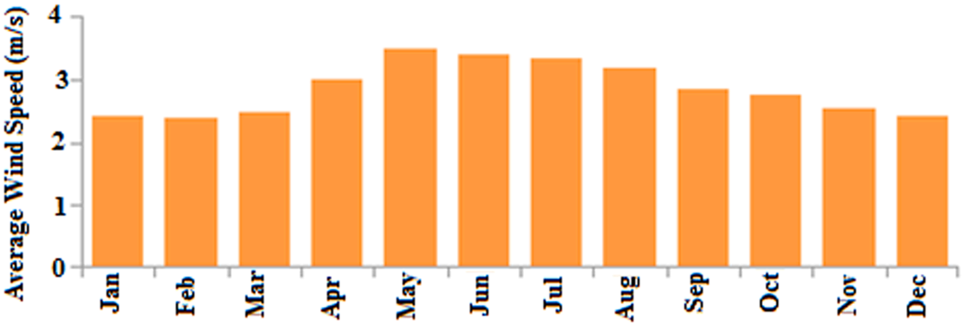

Fig. 4. Wind potential of Goma, obtained from SODA web site [15].

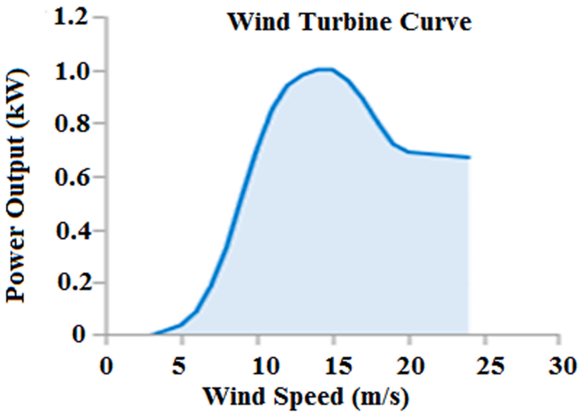

Fig. 5. Typical wind Turbine power curve [16].

possible to have microgrids that have PV systems integration, such as 3681 kW, 7269 kW and 1151 kW respectively for microgrid_1, microgrid_3, and microgrid_4, and 12,101 kW for the overall multi-microgrids system (Table 1).

Optimization results of each individual microgrid configuration using homer grid

The results of optimization of each microgrid are presented in Figs. 11–13.

In each of these configurations, the most economic case is compared to the base case. The optimization was done considering technical and economic aspects. The objective function was to minimize the net present cost evaluated at half life time of the project which is 12.5 years (based on the lifespan of the photovoltaic panel).

Fig. 11 presents the optimization results of microgrid_1 connected to the national grid.

The base case architecture here contains 3681 kW installed PV, 5696 kWh Batteries and 1 695 kW electronic converter connected to the grid. This system needs an initial capital of 8530,000£ and will have a net present cost (NPC) of 12,100,000£ after 12.5 years of its installation. Its operational and maintenance cost is 274,394£ per year and the levelized cost of energy (LCOE) is 0.14 £ per kWh.

However, Homer Grid optimization indicates that the base case is not actually the lowest cost system. Indeed, by removing the PV from the system and increasing slightly the size of the storage element (from 5696 kWh to 5707 kWh) the initial capital is significantly reduced (from 8530,000 £ to 2640,000 £) while the NPC stay practically the same (12,100,000£ and 12,000,000 £).

It is therefore important to observe that although the initial capital is considerably low for the system without PV, due to the increase in the operational and maintenance introduced by the absence of PV (from 274 394 £ per year to 721 083 £ per year), the total lifetime cost of this system is higher. This situation is quantified by the LCOE of this system which is 0.153 £ per kWh versus 0.14 £ per kWh for the system containing PV. For the lowest cost system, the annualized utility bill saving is – 456,540 £, the net present utility bill saving is – 5900,000 £ and the annualized demand charge saving is – 456,540 £.

Fig. 12 presents the optimization results of microgrid_1 connected to the Virunga private grid. These results indicate that for a base case architecture with 5707 kWh batteries and 1128 kW electronic converter connected to the Virunga private grid results in a winning system architecture composed of 15,357 kW installed PV, 5761 kWh batteries and 5907 kW. The base case system needs an initial capital of 2640,000 £ and will have a net present cost (NPC) of 19 800 000 £ after 12.5 years of its installation. Its operational and maintenance cost is 1330,000 £ per year and a levelized cost of energy (LCOE) of 0.253 £ per kWh. Homer Grid optimization indicates that the lowest cost system compared to the base case requires a high initial capital (25,500,000 £ versus 2640,000 £) while the NPC is significantly reduced (2480,000 £ versus 19,800,000 £). Due to its negative operational and maintenance cost (– 1780,000 £ per year),

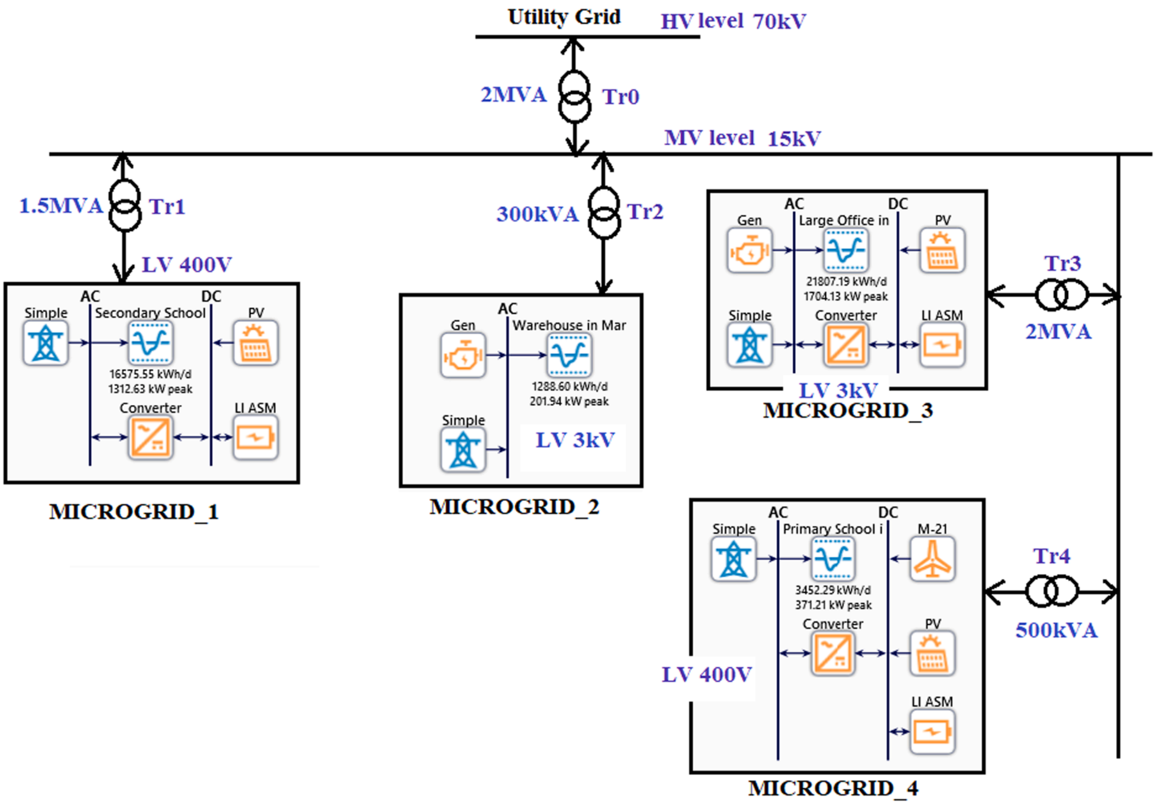

Fig. 6. Multi-microgrids simplified architecture.

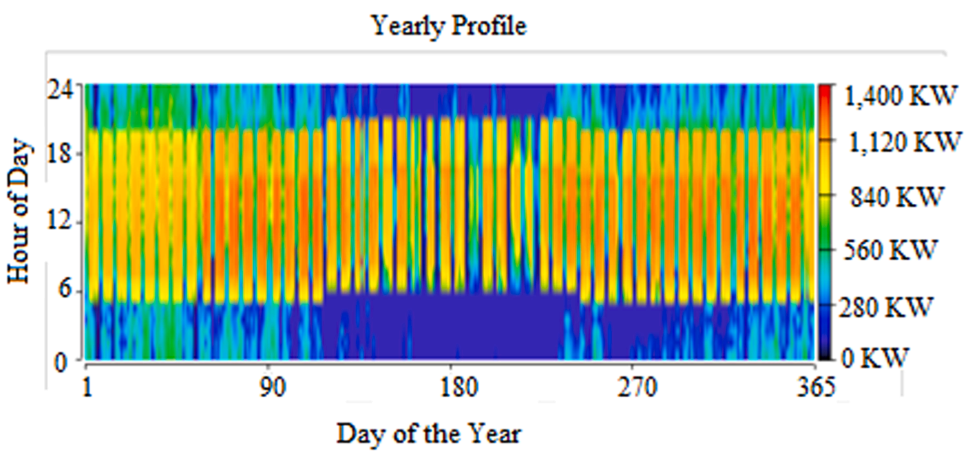

Fig. 7. Load Profile of microgrid_1.

the lowest cost system has a lower total lifetime cost. This situation is quantified by the LCOE of this system which is 0.0113 £ per kWh versus 0.14 £ per kWh for the base case system. After optimization, the annual utility bill saving is 3230,000 £, the net present utility bill saving is 41,700,000 £ and the annualized demand charge saving is 3230,000 £. The winning system presents a payback period of about 7 years. This period is calculated and compared to the base case as shown on the graph of Fig. 12.

Fig. 13 presents the optimization results of microgrid_3 connected to the national grid. This system presents a base case architecture with 1800 kW installed Gas Generator, 7269 kW installed PV, 8898 kWh batteries and 2557 kW electronic converter connected to the national grid. While the winning system architecture has only the gas generator. The base system needs an initial capital of 17,500,000 £ and will have a net present cost (NPC) of 20, 100,000 £. The operational and maintenance cost is 200,999 £ per year and the levelized cost of energy (LCOE) is 0.161 £ per kWh. The lowest cost system containing only the generator requires an initial capital of 12,700,000

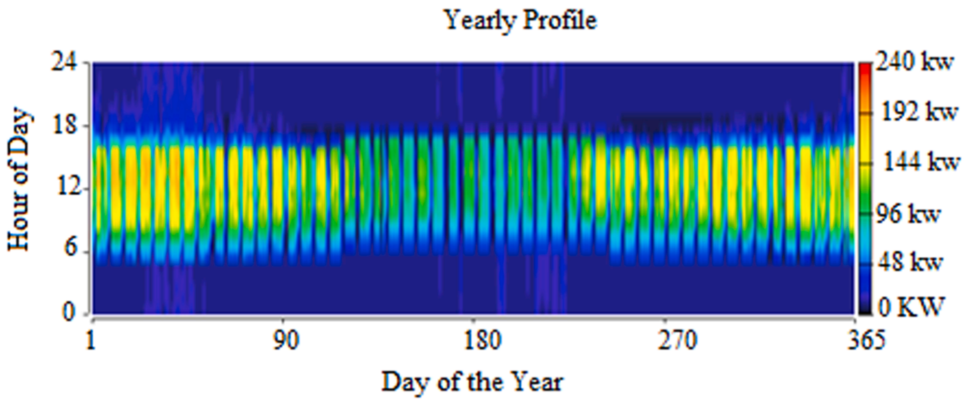

Fig. 8. Load Profile of microgrid_2.

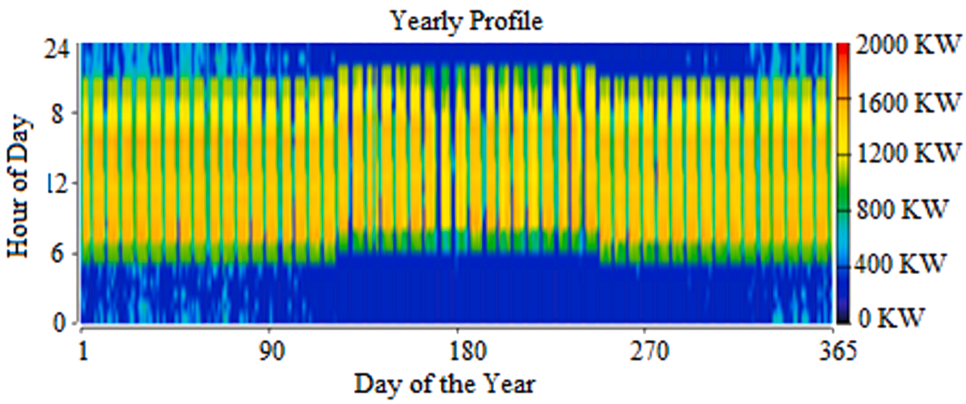

Fig. 9. Load Profile of microgrid_3.

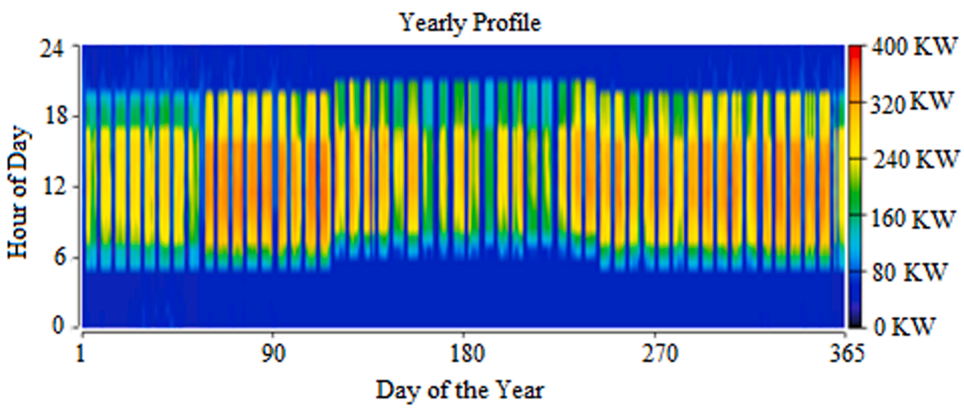

Fig. 10. Load Profile of microgrid_4.

£. The NPC is significantly reduced (2700,000 £ versus 200,999 £) but the operational and maintenance costs is higher (771,012 £ per year versus 200,999 £ per year), The LCOE of this system which is 0.123 £ per kWh versus 0.161 £ per kWh for the base case system.

Conclusion

This work investigates the impact of several microgrids' interconnection on reliability and power stability. This multi-microgrid was formed by interconnecting together four microgrids in Goma town, each one containing different types of elements (loads, energy sources, energy storage devices) in other to be able to see the impact of different type of renewable resources on the studied

Table 1
Summarized Optimization results of the overall system.

		Microgrid1	Microgrid2	Microgrid3	Microgrid4	Multi-microgrids
Load	Daily average energy (kWh)	16 575.55	1 288.6	21 807.19	3 452.29	43 123.63
	Peak power (kW)	1 312.63	201.94	1 704.13	371.21	3589.91
PV installed power (kW)		3 681		7 269	1 151	12 101
Generator rated power (kW)			210	1 800		2 010
Storage rated energy (kWh)		5 696		8 898	1 409	16 003
Converter rated power (kVA)		1 675		2 557	371	4 603

N.B.: Empty cells on the Table 1 above to parameters of microgrid not considered in the corresponding multi-microgrids configuration.

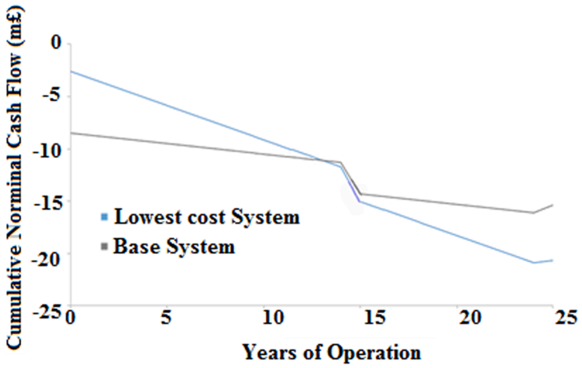

Fig. 11. Optimization results of microgrid_1 connected to the national grid.

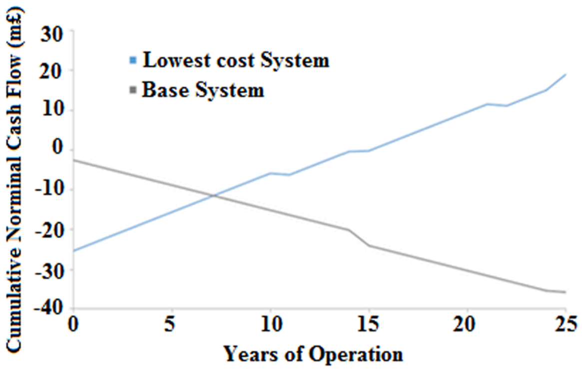

Fig. 12. Optimization results of microgrid_1 connected to the Virunga private grid.

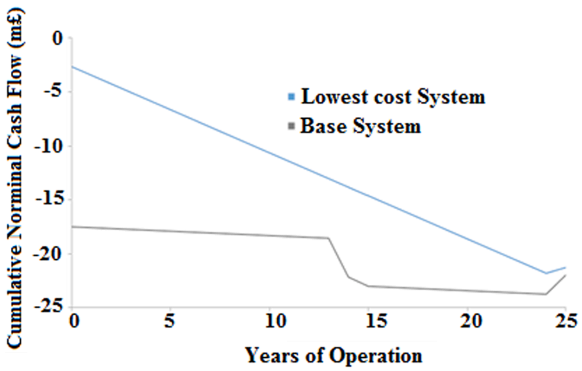

Fig. 13. Optimization results of microgrid_3 connected to the national grid.

parameters. From the potential assessment of the case study, it was observed that methane gas present in Kivu Lake and solar PV have higher potentials. On the other hand, when connected to the Virunga private grid, the system with PV can be sold to the grid with a payback period of about seven years eventhough it requires a higher initial capital. Hence, the interconnection of several microgrids have the potential to yield a more stable and reliable system when integrating different DG units.

CRediT authorship contribution statement

Divine Khan Ngwashi: Conceptualization, Investigation, Methodology, Writing – original draft. **Arsene Kafatiya Arnold:** Conceptualization, Methodology, Writing – review & editing. **Shu Godwill Ndeh:** Writing – review & editing. **Emmanuel Tanyi:** Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

This work was supported by the Faculty of Engineering and Technology of the University of Buea, Cameroon.

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